

# Project Scope & Objectives Document

**Project Name:** AI Job Butler - Personal Job Search & Application Assistant <sup>1</sup>

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## 1. Project Overview

The AI Job Butler is a personal, intelligent job-search assistant designed to automate and optimize the job-hunting process<sup>3</sup>. It will know the user's profile, preferences, and job history like a personal butler, and will: continuously monitor job portals, tailor resumes & cover letters dynamically, apply to jobs automatically (or with approval), send cold emails to recruiters, and maintain a dashboard showing job application status and metrics<sup>4</sup>.

The system runs locally on a Ryzen 9 laptop and optionally integrates with a Streamlit-hosted profile dashboard<sup>5</sup>.

## 2. Objectives

- **Automate Job Search & Application** with human approval <sup>6</sup>
- **Cold Outreach** with follow-up tracking <sup>7</sup>
- **Resume & Cover Letter tailoring** <sup>8</sup>
- **Human-in-the-Loop Approval Queue** for reviewing drafts <sup>9</sup>
- **Centralized Dashboard** for status tracking <sup>10</sup>
- **Data Privacy & Control** with local storage <sup>11</sup>

## 3. Key Features

- **Job Scout** (scrapes portals & stores data) <sup>12</sup>
- **Profile Memory** (vector DB with skills & history) <sup>13</sup>
- **Resume Tailor** (LLM-powered) <sup>14</sup>
- **Approval Queue** (manual review before sending) <sup>15</sup>
- **Auto-Apply Engine** (Playwright automation) <sup>16</sup>
- **Cold Mailer** with Follow-ups <sup>17</sup>
- **Rate Limiter** for safe scraping & emailing <sup>18</sup>
- **Streamlit Dashboard** (applications, chat, metrics) <sup>19</sup>

## 4. Technology Stack

- **Frontend:** Streamlit <sup>20</sup>
- **Backend:** FastAPI <sup>21</sup>
- **Orchestration:** LangChain + LangGraph <sup>22</sup>
- **Database:** SQLite/Postgres <sup>23</sup>
- **Memory:** ChromaDB <sup>24</sup>
- **LLM:** Ollama (Llama3/Mistral) or OpenAI API <sup>25</sup>
- **Automation:** Playwright <sup>26</sup>
- **Email:** Gmail API <sup>27</sup>
- **Scheduler:** APScheduler <sup>28</sup>
- **Rate Limiting:** Configurable delays + jitter for human-like behavior <sup>29</sup>

## 5. Risks & Mitigation

- **Website changes** → Modular scrapers <sup>30</sup>
- **Spam risk** → Gmail API + delays <sup>31</sup>
- **Duplicate applications** → History tracking <sup>32</sup>
- **LLM hallucinations** → Approval queue validation <sup>33</sup>
- **Data loss** → Local & encrypted cloud backup <sup>34</sup>

## 6. Future Enhancements

- Recruiter-side dashboard <sup>35</sup>
- Interview prep agent <sup>36</sup>
- Networking automation <sup>37</sup>
- Analytics (success rates by portal) <sup>38</sup>
- Full offline mode with local embedding <sup>39</sup>